Ravi Gyani

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Quick Summary

- 20+ years in software engineering, specializing Cloud Computing, **Serverless solutions** and Big Data.
- Reputed as a **Problem Solver**, bringing out-of-box approaches to system design
- Focused on **optimizing performance** and designing **cost-effective** architectures.
- Advocate for validating ideas via **proof-of-concepts** (POCs).
- Firm believer that knowledge shared is knowledge squared.

Technologies

- **AWS Expertise:** IAM, EC2, S3, Lambda, Step Functions, Batch, API Gateway, AppSync, DynamoDB, IoT, EventBridge, CloudWatch, RDS, Aurora, Redshift, Glue, Athena, ECS, Elastic Beanstalk, CloudFront, Route 53, ECR, EMR, Kinesis, Greengrass and more.
- **Infrastructure-as-Code**: Docker, CDK, Terraform, CloudFormation
- **Big Data:** Hadoop Ecosystem, HBase, Kafka, Spark, Cassandra, Redis
- **Programming Languages:** Python, Java, TypeScript, Go, Node.js

Certifications

- AWS Certified Security Specialty (SCS-C01) - Mar 2023
- **AWS Certified Solutions Architect Professional (SAP-C01)** - Oct 2020
- AWS Certified Solutions Architect Associate (SAP-C02) – Oct 2020
- Databricks Certified Developer: Apache Spark 2.X – Jul 2019
- Confluent Certified Developer for Apache Kafka – Jul 2019

Education

Bachelors of Engineering (Netaji Subhas Institute of Technology, Delhi) 1999-2003

Work Experience

Senior Solutions Architect at Luxoft GmbH, Munich, Jan 2021 – July 2024
Consulted for customers such as Amazon, BMW, Continental and Volkswagen**.**

Key Responsibilities and Achievements:

- **Cloud Solutions and Architecture:**
  - Designed, engineered, deployed **Serverless** architectures using Python, Java, CDK and AWS services, ensuring high availability and scalability.
  - Re-platformed legacy applications to AWS, modernizing infrastructure and reducing operational overhead.
  - **Optimized** mission-critical DynamoDB tables with direct API Gateway integrations, reducing response times to 10ms.
  - Conducted architecture reviews and AWS "Well-Architected Reviews", providing actionable insights to enhance systems' performance and reliability.
- **Automation and Cost Optimization:**
  - **Automated repetitive tasks**, such as custom AMI generation using AWS Step Functions, improving developer productivity and minimizing errors.
  - **Redesigned** a batch computing service using Airflow, Batch, Docker, and EFS, **achieving a 90% cost reduction** by leveraging S3 runtime mounts, Step Functions, and Spot Instances.
  - Streamlined multi-account/multi-environment code deployments with CDK and AWS CodePipeline, ensuring faster and error-free delivery cycles.
- **Big Data and Machine Learning:**
  - Migrated a legacy GCP application to AWS, implementing ETL pipelines to process and prepare data. Applied machine learning algorithms to derive actionable metrics from incoming data streams.
  - Implemented **machine learning federated learning solutions** for fault detection using AWS IoT Core, ECS, and greengrass.
- **Innovation in Solutions and Tools:**
  - Designed a custom in-vehicle support solutions leveraging AppSync and Chime for real-time communication and support.
  - Built a custom Grafana plugin (Go and Angular) to query MapR clusters via Drill REST API, enhancing data analysis workflows.
  - Developed a custom Zeppelin interpreter (Java and Angular) for Git interactions, solving critical development challenges.
  - Engineered the migration of a legacy application to event-driven architecture with EventBridge, facilitating real-time event processing, achieving improved scalability and reduced dependency bottlenecks.
  - **Engaged in presales processes**, including solution design, POC development, and cost/time estimation.
- **Leadership and Training:**
  - Delivered IaC and Serverless workshops for teams, focusing on CDK, and Terraform.
  - Guided teams through knowledge-sharing sessions, fostering skill development and promoting innovation.

Solutions Architect at Polaris Wireless, Bangalore, Feb 2014 – Aug 2020
Designed, Engineered, Delivered an innovative scalable big data solution

Key Responsibilities and Achievements:

- Initially joined as an iOS/Android Tech Lead, identified scalability issues with the company's SQL Server-based product for data storage and processing. Transitioned into a Big Data Developer role to address these challenges.
- Designed and developed the first **proof of concept (POC)** for a scalable Big Data solution, successfully presenting the idea to both management and clients, and oversaw its first deployment on-site.
- **Big Data Solution Development**:
    - Built a bare-metal Big Data solution to store and profile mobile location data for all subscribers of a country, tailored for use by investigative agencies.
    - Leveraged technologies such as HBase, Cassandra, MongoDB, Solr, Kafka, and Spark.
    - Developed the entire product using Java, Scala, and Python, incorporating KPIs and benchmarking for ingestion and performance.
    - Led a team of nine engineers, ensuring efficient development, deployment, and delivery.
- **Microservices and Algorithms**:
    - Designed and implemented microservices to generate profiles and algorithms for comparison of profiles in near-real time.
    - Algorithms identified familial, communication, and location patterns among subscribers, **earning recognition as a named contributor on a patent**.
    - Optimized the system using distributed computing technologies, including MapReduce and Spark, with HBase over Hadoop for storage and Kafka as the communication backbone.
- **Next-Generation Solution:**
    - Pioneered a new version of the product with in-line profiling during data ingestion via Apache Spark
    - Leveraged AWS services such as S3 for storage, EMR for processing large-scale data, and MSK for ingestion pipelines, driving the evolution of the next-generation product
    - Integrated advanced cardinality solutions (Count-Min Sketch, Bloom Filters, HyperLogLog) for efficient data profiling and storage optimization.

iOS and Android, C# Winforms developer, 2003-2014
Worked for various companies

Founded KNOCKS Solutions (2006- 2009) developing third party .NET Winforms Components Suite, like Outlook style Calendar, Visual Studio like Docking Extender, Office style Ribbon Control, Winamp like Skin-extender and various meters and gauges UI components